

JET TRANSFORMATION BLUEPRINT

Smart-Lean Bio-Thermal Infrastructure

LWANDILE ENGINEERING PROJECTS

Phase I

The Core Strategy

Biological Security & Thermal Stability

Three Pillars of Asset Defense



Theory of Unwelcome

Structural firewalls cap biological asset exposure at 25% by isolating airborne pathogens completely.



BESS Sovereignty

30kW/30kWh off-grid shield ensuring zero thermal shock cycles during highveld winter fronts.



Time Rotation

Maximizing revenue velocity via high-occupancy pod rotation across separate specialized structures.

The Poultry Sprint Engine

5,760 Bird Capacity

- ✓ 384 m² Open-Sided House (12m x 32m)
- ✓ Passive Convective Siphon Ridge Design
- ✓ 50mm High-Density Roof Insulation Boards
- ✓ Integrated 128 m² Layered Brooding Zone

Optimized to pull zero-watt ventilation baseloads from our energy architecture.



Storm & Rain Mitigation Protocol

Defending Open-Sided Assets

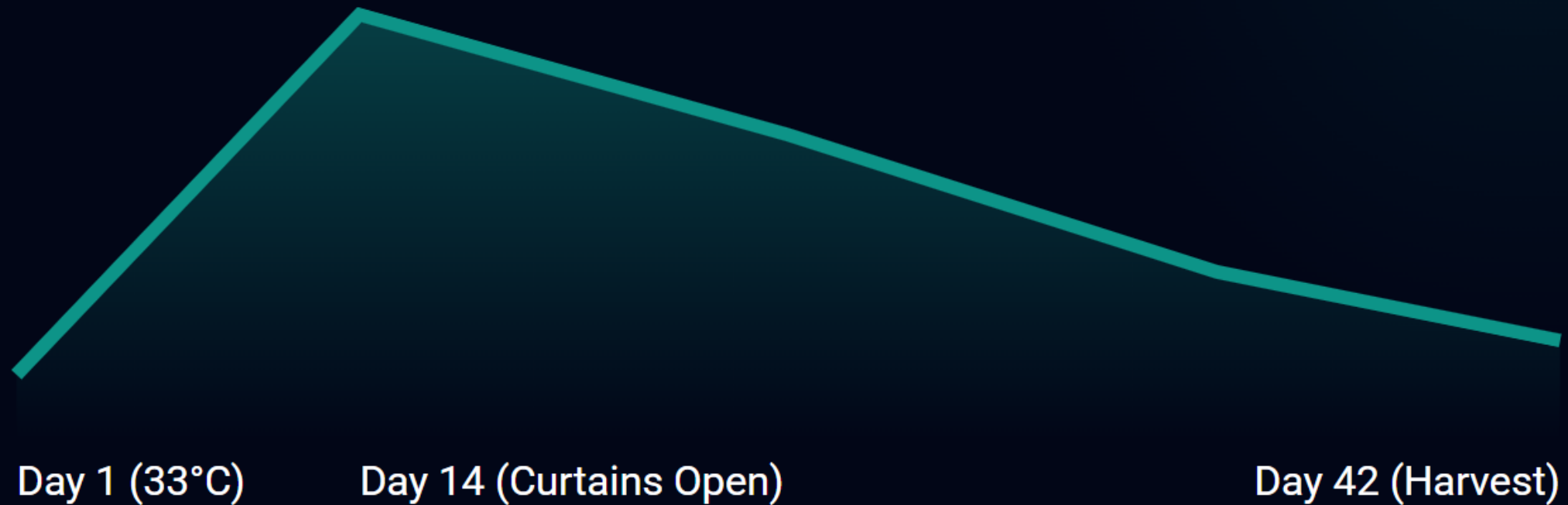
- ✓ **Winch Curtain Interlock:** Heavy 550g/m² PVC curtains lock down within 90 seconds to prevent moisture cross-spray.
- ✓ **600mm Dwarf Wall Buffer:** Plastered block base prevents floor-level rainwater ingress and protects ground-level nesting spaces.
- ✓ **Extended Eaves Overhang:** Generous roof eaves divert heavy downpours away from open ventilation slots



Thermal Integrity

Curtain adjustments seal weather fronts out while the ridge siphon maintains baseline oxygen exchange without stalling.

Thermal Transition Lifecycle



Surgical step-down microclimates protect BESS reserves after Day 14 canopy expansions.

The Swine Pipeline

4x12m Infantry Shield & 12x32m Main Finisher

The 4-Pod Nursery Fortress

Stage 1: Airborne Boundary

- ✓ 48 m² Standalone Masonry Footprint
- ✓ Double-Skin Cavity Wall Insulation Shield
- ✓ Four Independent 12 m² Bio-Pods
- ✓ Airtight Gaskets & Thermal Barriers

Isolates vulnerable weaners during their critical 6-week post-weaning immunity gap.

Sovereign Aster **Divided** Biosecure Pods



Immunological Firewall

Four physically sealed
3m x 4m nursery pods.
cap risk at 25% of the batch.

Four Utility Costs 🏠

Infections are isolated to each pod.

Biosecurity: 🛡️

Risk structurally limited to 25%.

Food & Energy composting between cycles.

Defensive Pod Engineering



Microclimates

Zonal carbon heating mats embedded directly into floor slats. Heat is focused only on active pods to conserve energy.



Hygienic Slats

Elevated plastic slatted flooring allows quick waste drops. Keeping the bed dry breaks structural pathogen lifecycles.



Airtight Seal

Vapor-tight aluminum exterior doors eliminate cross-contamination drafts between nursery chambers completely.

Capacity & Spatial Allocation

Asset Unit	Footprint Dimensions	Target Capacity	Climate Target
Nursery Pod (x4)	48 m ² Total Block	160 Weaners / Batch	32°C Base Control
Finisher Barn	384 m ² (12m x 32m)	160 Hogs / Batch	18°C - 20°C Ambient
Poultry Barn	384 m ² (12m x 32m)	5,760 Birds / Batch	Dynamic Step-Down

Annual Throughput Payload

480

Hogs Per Annum

The Staggered Math

160 Weaners per Batch x 3 Rotations per Year =
480 Market Hogs.

Nursery Residence Time: 6 Weeks.

Finisher Residence Time: 16 Weeks.

Prevents pipeline overlap collisions, matching
facility limits seamlessly.

Continuous Flow Pipeline



Wave 1 (January)

Nursery Pods filled.
BESS microclimate active.

Wave 2 (May)

Wave 1 transitions to
Finisher.
Nursery sanitized &
reloaded.

Wave 3 (September)

Continuous pipeline
occupancy.
Steady-state cash velocity.

STEWARDSHIP LOCKED

Defensive. Lean. Investment-Ready.

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Mpumalanga | JET Transformation Framework

JET TRANSFORMATION BLUEPRINT

Phase II: Advanced Operational Lifelines

Integrating Hydrology, Biosecurity, Gravity Logistics & IoT Dashboard Systems

The Complete Industrial Ecosystem

A truly optimized bio-thermal engine cannot rely solely on structural walls. It requires a flawless supply chain of critical inputs and waste extraction mechanics to protect core livestock capitals.



Hydrological Engine

Ensuring infinite water availability, sanitation dosing, and processing slurry volumes completely off-grid.



Gravity Logistics

Utilizing structural elevations and passive gradients to evacuate waste streams directly into biogas digestors.

The Hydrological Engine: Sizing

Water Security Mandate

Livestock assets collapse within hours of hydration failure. Our system designs a multi-day buffer framework to absorb extreme grid or equipment disruptions.

- ✓ **Daily Demand Peak:** Cross-calculated for 5,760 broilers and 160 active hogs simultaneously.
- ✓ **Storage Threshold:** 5-Day autonomous water reserve buffer held in an elevated tower network.
- ✓ **Digester Integration:** Consistent flow controls supply moisture for the active organic anaerobic slurry.

“A”

Pumping Matrix

Dedicated low-inductive solar-borehole pumps run exclusively on daytime solar direct currents, completely bypassing battery reserves during nighttime operational cycles.

Water Purification & Sanitization

Purification Stage	Technology Deployed	Target Vector Eliminate	BESS Power Profile
Primary Phase	50-Micron Sediment Disk Filter	Silt, Sand, and Particulate Matter	Zero-Watt Mechanical Flow
Biological Shield	Inline Ultraviolet (UV) Sterilizer	Bacterial Pathogens & Waterborne Viruses	Low-Voltage DC Current
Residual Defense	Proportional Auto-Chlorination	Biofilm Accumulation in Drinker Lines	Zero-Watt Kinetic Dosing

Gravity Slurry Topography

Eliminating Mechanical Pumps

Pumping raw hog manure draws heavy electrical currents and invites constant mechanical blockages. We use earth physics to bypass this bottleneck entirely.

- ✓ **The 1-in-50 Gradient Rule:** All sub-slab concrete flush channels are poured with a strict 2% drop toward the primary collection sump.
- ✓ **Scrape-Free Separation:** Moisture and solid organic fractions move continuously via hydrostatic gravity displacement.



Digester Intake Coupling

The collection sump sits at the lowest topographical point on site, allowing automated hydraulic head pressure to push slurry directly into the digester intake valve.

Biosecurity Access Barriers

Vehicle Perimeter Gate

Forced spray-dip boot architecture at the main perimeter line. Neutralizes pathogenic spores tracked on truck tires before offloading feed components.



Shower-In, Shower-Out

Dedicated change-house locker system. Workers change from external garments into sanitized, farm-allocated uniforms before approaching animal units.



Localized Pod Baths

Deep chemical walk-through footbaths placed at the entry thresholds of every single nursery pod to stop boot-borne cross-contamination.

The IoT Sensor Nervous System

Data-Driven Stewardship

Precision management requires real-time insight. Low-power IoT sensors form a continuous monitoring grid, sending critical alerts directly to the digital dashboard.

- ✓ **Sub-Second Resolution:** Tracks room environments constantly without needing manual clipboard checks.
- ✓ **Predictive Analytics:** Catches structural deviations before they manifest as biological stress or clinical disease.



LoRaWAN Protocol

Sensors communicate over long-range, ultra-low-power radio frequencies, ensuring zero interference with farm operations and minimal electrical draw on the BESS.

Nursery Pod Microclimate Tracking

Sensor Matrix Type	Target Threshold Limit	Automated Logic Action Triggered	Critical Asset Protected
Ambient Temperature	Drops below 30°C in Nursery	Modulates Biogas Steam Valves Open	Piglet Digestion & Vitality
Ammonia (NH3) Gas	Exceeds 20 Parts Per Million (PPM)	Triggers Linear Curtain Winch Opening	Respiratory Mucosal Linings
Relative Humidity	Exceeds 75% Moisture Saturation	Activates Roof Siphon Damper Adjustments	Pathogen Spore Supression

Inline Hydrological Monitoring

The Consumption Early-Warning System

Animals display a drop in water intake up to 48 hours before visible clinical symptoms of illness appear. Digital flow meters exploit this biological trait.

- ✓ **Pulse-Output Flow Meters:** Installed on the main supply lines of each independent nursery pod and poultry section.
- ✓ **Algorithmic Deviation Alerts:** Flags a 10% drop against historical baseline consumption patterns immediately on the dashboard.



Surgical Isolation

Enables immediate veterinary intervention on a specific pod, preventing blanket medication costs across healthy batches.

Automated Feed Logistics



Bulk Silo Load-Cells

Digital strain-gauge load-cells mounted to silo legs weigh feed stocks continuously, flagging automatic reorder points directly to suppliers.



Enclosed Augering

Flex-auger lines move feed from outside storage straight into internal house hoppers, eliminating bag handling and rodent contamination vectors.



Real-Time FCR Tracking

Correlating daily weight metrics against automated feed dispense rates to calculate the Feed Conversion Ratio instantly on the dashboard.

The Co-Generation Integration Matrix

Synthesizing our primary lifelines into a cohesive waste-to-energy infrastructure. Every animal byproduct serves as a fuel input for environmental defense.

Input Source	Processing Unit	Co-Generated Derivative	Operational Destination
Gravity-Fed Pig Slurry	Anaerobic Digester	Methane Rich Biogas	Internal Fuel Storage Tanks
Combused Methane Fuel	Generator Set (GenSet)	Electrical Volts (AC/DC)	30kW/30kWh BESS Storage Bank
Engine Thermal Exhaust	Heat Exchanger Jacket	High-Temperature Steam Heat	Sub-Floor Radiant Slab Lines

The Thermal Balancing Logic

Prioritizing Energy Allocation

Our automation system applies strict hierarchical logic to handle environmental loads without over-stressing structural energy boundaries.

- ✓ **Primary Heat Load:** Handled entirely by the biogas-driven sub-surface radiant steam lines during cold nights.
- ✓ **Secondary Heat Load:** Backup electric carbon-fiber mats activate only if steam temperatures drop below set parameters.



BESS Preservation

By using direct thermal steam for base heating, battery banks stay reserved for critical electronics, lighting, and ventilation controls.

Climatic Resilience Parameters



Winter Frost Armor

Insulated cavity walls and floor-level steam heat combine to withstand deep South African highveld winter sub-zero drops effortlessly.



Summer Heat Shedding

Passive ridge chimneys and wide roof overhangs vent extreme radiant loads naturally without requiring high-amp mechanical cooling.

Funder Compliance Verification

JET Alignment Matrix

This design satisfies the complex green economy asset criteria mandated by international investment consortiums.

- ✓ **Carbon Reductions:** Direct methane capture via digestion avoids standard manure greenhouse venting.
- ✓ **Resource Efficiency:** Zero grid reliance and a closed-loop water sanitization infrastructure.



Bankable Asset Model

By engineering out climate, energy, and biological risks, the plan transforms volatile farming into a predictable industrial pipeline. The Sovereign Food Energy Artifact.

OPERATIONAL ECOSYSTEM COMPLETE

**Infrastructure. Automation. Absolute
Precision.**

Lwandile Engineering Projects (Pty) Ltd

Precision Alliance Architecture | Mpumalanga